Saad Alif A Zaman

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Research Interests

● **Data-Efficient Deep Learning Architectures:** Developing systematic cross-validation protocols, mathematically optimized augmentations, and instance segmentation frameworks to overcome extreme data scarcity in highly dynamic, unstructured field environments.
● **Vision-Based Environmental & Hydrodynamic Modeling:** Extracting high-fidelity physical metrics, such as fluid-structure interaction, wave kinematics, and environmental anomalies, from heterogeneous surface camera feeds to advance AI-powered predictive modeling.
● **Automated Visual Inspection (AVI) & Condition Assessment:** Translating static vision models into robust offline perception pipelines for the non-contact, in-situ structural health monitoring of heavy civil and industrial infrastructure.

Education

B.Sc. In Naval Architecture And Marine Engineering (NAME) (April 2019 – July 2024)
Bangladesh University of Engineering and Technology (BUET)
Cumulative GPA: 3.37 / 4.00

Relevant Coursework:
● **Computation & Mathematics:** Computer Programming I & II (C/C++, FORTRAN, OOP) [A+]; Statistics, PDEs, and Matrices (Applied Probability & Linear Algebra) [A]; Numerical Computations [B]
● **Structural Health & Degradation:** Marine Maintenance and Repair (Corrosion, Fatigue, Biofouling & Defect Analysis) [A-]; Welding Technology (NDT & Defect Inspection) [B+]; Finite Element Method for Ship Structure [B+]
● **Mechanics, Materials & Dynamics:** Mechanics of Structure [A]; Shipbuilding Materials [B+]; Ship Vibration (Free/Forced Multi-Degree Systems) [B+]
● **Hydrodynamics & Environmental Physics:** Fluid Mechanics [S]; Marine Hydrodynamics [B]; Heat Transfer (Steady/Transient Conduction) [B]

Research Experience

➢ **Independent Research Project** (August 2025 – Present)
"Interaction-Effect Metric for Data-Scarce Industrial Vision: Application to Ship Hull Biofouling Inspection"
Research advisors: Dr. Kazi Naimul Hoque, Associate Professor, Dept. of Naval Architecture and Marine Engineering, BUET
Samiul Based Shuvo, Assistant Professor, Dept. of Biomedical Engineering, BUET

● **Engineered** a leakage-free cross-validation protocol (k-n Fold ACV) to reduce test-train data split bias, alongside an Interaction Effect Metric to optimize augmentation pipelines for instance segmentation architectures (YOLOv8/11, Mask R-CNN).
● **Validated** the framework on a newly curated field dataset (NDBD, n=35), overcoming the extreme visual data scarcity of noisy operational environments to extract reliable physical measurements from limited imagery.
● **Achieved** a 47% mAP improvement over baseline training, executing cross-model validation and making the full codebase and dataset publicly available to ensure reproducibility.

➢ **Undergraduate Thesis** Final Grade: A+ (4.00/4.00) (June 2023 – July 2024)
"Developing a biofouling & corrosion identification and distribution system using video images based on deep learning"
Supervisor: Dr. Kazi Naimul Hoque, Associate Professor, Dept of Naval Architecture and Marine Engineering, BUET

● **Collected and curated** in-situ field data to build the Narayanganj Dockyard Biofouling Dataset (NDBD) directly from Dockyard & Engineering Works Ltd., Narayanganj (publicly available on GitHub), supplementing it with public bridge inspection data for steel corrosion severity.
● **Deployed** instance segmentation pipelines on heterogeneous, real-world structural data, demonstrating high-fidelity feature extraction that can be generalized to broader civil infrastructure monitoring, environmental material degradation, and automated condition assessment.
● **Developed** a video-based detection prototype, establishing the foundational perception pipeline required to integrate robust vision models with autonomous platforms (e.g., UAVs and ROVs) for continuous structural monitoring.
● **Secured** the competitive RISE Student Research Grant to fund the system's development.

➢ **Junior Year Project** Final Grade: A+ (4.00/4.00) (May 2022 – May 2023)
"Designing an Inland 100 Passenger Ship for Inland waterways of Bangladesh"
Supervisor: Dr. Md. Mashiur Rahaman, Professor, Dept of Naval Architecture and Marine Engineering, BUET

● **Designed** a conceptual 100-passenger inland vessel with an electric propulsion system tailored to Bangladesh's inland waterways.
● **Optimized** the hull design and executed systems-level optimization of mechanical networks, utilizing finite element validation to resolve complex structural constraints and ensure physical integrity under dynamic loads.
● **Improved** energy efficiency compared to conventional diesel systems.

Publications

1. Minhazul Islam, **Saad Alif A Zaman**, Md. Mashiur Rahaman, "Concept of an Electric-Powered Passenger Vessel in Bangladesh," Journal of Maritime Research, vol. 20, no. 3, pp. 68–71, Dec. 2023

2. **Saad Alif A Zaman**, Samiul Based Shuvo, Kazi Naimul Hoque, "Interaction-Effect Metric for Data-Scarce Industrial Vision: Application to Ship Hull Biofouling Inspection," Preprint; in revision following peer review, targeting resubmission, 2026. [SSRN link]

RESEARCH GRANTS, RECOGNITION & SCHOLARSHIPS

- **RISE Student Research Grant 2023:**
  Awarded a competitive Student Research Grant (BDT 65,790) by **R**esearch and **I**nnovation center for **S**cience and **E**ngineering (RISE) for the research project: "*Developing a biofouling & corrosion identification and distribution system using video images based on deep learning.*"

- **SPARRSO Research Fellowship Program 2025–26:**
  Shortlisted (final round) for a national research fellowship with the Space Research and Remote Sensing Organization (SPARRSO), Bangladesh; invited to present the proposal "*Development of a **Fixed-Wing VTOL Quad Plane**,"* proposing an autonomous aerial vision platform specifically engineered for non-contact optical inspection and rapid post-event reconnaissance of large-scale infrastructure.

- **University Stipend Scholarship-2019, 2020, 2021 – BUET:**

- **Rajshahi Board Scholarship HSC (Grade 12 equivalent) 2018:**

STANDARDIZED TEST SCORES

International English Language Testing System (IELTS) Academic (15th December, 2025)
Overall Band Score: 7.5 (Listening: 8.5, Reading: 8.5, Writing: 6.5, Speaking: 7.0)

TEACHING & ACADEMIC EXPERIENCE

Materials Development Instructor (April 2025 – August 2025)
Udvash Academic & Admission Care | Dhaka, Bangladesh
● **Evaluated** question quality, solution accuracy, and curriculum alignment for high-stakes engineering and university admission examinations.
● **Developed and reviewed** academic content, rigorous assessment materials and model tests for grade 10-12 and pre-university levels.

Academic Mentor (Mathematics & Physics) (2019 – Present)
● **Instruct** secondary and higher-secondary students in foundational and advanced mathematics and physics through targeted small-batch and one-to-one sessions.
● **Guide** students through complex problem-solving methodologies across both English and Bangla medium curricula, leading to multiple successful admissions to top-tier engineering institutions including BUET, KUET, DU, and IUT.

Industrial Internship – Shipyard Practice (October 2022 – November 2022)
Khulna Shipyard Limited | Khulna, Bangladesh
● **Industrial Fabrication & Manufacturing:** Gained direct exposure to heavy-scale industrial manufacturing processes, including CNC machining, structural welding, and multi-stage sub-assembly, bridging theoretical design with real-world physical construction.
● **Structural Design & Systems Integration:** Analyzed complex hull, piping, and equipment networks, developing a practical understanding of structural geometries, load constraints, and field assembly operations.
● **Field Deployment Context:** Navigated active shipyard environments (from drydock erection to delivery trials), gaining the hands-on industrial intuition required to deploy practical vision-based SHM and NDT systems in dynamic, unstructured field settings.

TECHNICAL SKILLS

● **Programming:** Python, C, C++
● **Deep Learning**: Ultralytics YOLO, Mask R-CNN, PyTorch, TensorFlow
● **Data Annotation & Processing**: CVAT, Roboflow, LabelImg
● **Research Computing & Collaboration**: Google Colab, Kaggle
● **CAD software**: AutoCAD, Rhino
● **Finite Element Analysis software:** Abaqus
● **Document Preparation**: LaTeX, Prism, Microsoft Office

REFERENCES

Dr. Kazi Naimul Hoque
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and Research Advisor
Dept. of Naval Architecture and Marine Engineering,
BUET, Dhaka-1000, Bangladesh
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Faculty Profile

Samiul Based Shuvo
Assistant Professor, Research Advisor
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Faculty Profile

Dr. Md. Mashud Karim
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Faculty Profile